

Aer Estimator wrongly reads global phase

<https://github.com/Qiskit/qiskit-aer/issues/1795>

ROW 92

Aer Estimator wrongly reads global phase #1795

[New issue](#)[Closed](#)

A-ta-v opened on Apr 26, 2023

...

Informations

- Qiskit Aer version: 0.12.0
- Python version: 3.10
- Operating system: Mac OS Ventura, Intel

What is the current behavior?

Aer Estimator crashes while running simulation.

Steps to reproduce the problem

Script below

What is the expected behavior?

I create a parameter placeholder at the beginning.
After prepending the circuit with ZFeatureMap and several compositions the final circuit obtains a non-zero global phase represented as a linear combination of placeholder's entries. I do *NOT* change the global phase myself, rather this is the outcome of aforementioned operations.
At the time of evaluation, the linear combination of placeholder's entries cannot not be cast to a float value causing an exception.

In the simple example below, I reproduced the behaviour by explicitly expressing the global phase as a sum of two placeholder entries.
Note, the default estimator works without a problem.

Suggested solutions

Checking of global phase before evaluation is needed.

Script reproducing the issue:

```
import time
import numpy as np
from qiskit import QuantumCircuit
from qiskit.circuit import ParameterVector
from qiskit.primitives import Estimator as BackendEstimator
from qiskit.providers.fake_provider import FakeKolkataV2
from qiskit.quantum_info import SparsePauliOp
from qiskit.utils import algorithm_globals
from qiskit_aer.primitives import Estimator as AerEstimator

def report():
    num_qubits = 10
    algorithm_globals.random_seed = 1234567
    np.random.seed(algorithm_globals.random_seed)

    # Create a circuit with parameter placeholder.
    params = ParameterVector(name="r", length=num_qubits - 1)
    qc = QuantumCircuit(num_qubits)
    for i in range(num_qubits - 1):
        qc.cx(i, i + 1)
        qc.rz(params[i], i)
    qc.global_phase = params[0] + params[1] # <<<<< MAKES THE PROBLEM

    params = np.random.rand(len(params))
    observable = SparsePauliOp.from_list([("Z" * qc.num_qubits, 1)])
```

Assignees

No one assigned

Labels

bug

Type

No type

Projects

No projects

Milestone

Aer 0.12.1

Closed on Jun 15, 2023, 100% complete

Relationships

None yet

Development

No branches or pull requests

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Participants



```

tic = time.perf_counter()
backend_options = {"seed_simulator": algorithm_globals.random_seed, "shots": 4096}
use_aer = 1 # <<<< choose estimator here
if use_aer != 0:
    transpile_options = {"seed_transpiler": algorithm_globals.random_seed}
    run_options = {"shots": 1000}
    estimator = AerEstimator(
        backend_options=backend_options,
        transpile_options=transpile_options,
        run_options=run_options,
    )
else:
    estimator = BackendEstimator(FakeKolkataV2(), options=backend_options)
toc = time.perf_counter() - tic
print(f"Estimator created in {toc:.2f} seconds")

for i in range(20):
    tic = time.perf_counter()
    result = estimator.run(qc, observable, params).result()
    toc = time.perf_counter() - tic
    print(f"Attempt {i}, result obtained in {toc:.2f} seconds")
    print(f"Result: {result}\n-----\n")

report()

```

Script's output:

```

% py3 bug_report.py
Number of qubits: 10
Estimator created in 0.00 seconds
Traceback (most recent call last):
  File "/Users/username/venv/qu/lib/python3.10/site-packages/qiskit/circuit/parameterexpression.py", line 472,
    return float(self._symbol_expr)
  File "symengine_wrapper.pyx", line 1155, in symengine.lib.symengine_wrapper.Basic.__float__
  File "symengine_wrapper.pyx", line 980, in symengine.lib.symengine_wrapper.Basic.n
  File "symengine_wrapper.pyx", line 4383, in symengine.lib.symengine_wrapper.evalf
RuntimeError: Symbol cannot be evaluated.

```

The above exception was the direct cause of the following exception:

```

Traceback (most recent call last):
  File "/Users/username/work/qml_time_evol_research/bug_report.py", line 243, in <module>
    run_circuit_on_simulator(_num_qubits, _num_units, _num_layers, _insert_barriers)
  File "/Users/username/work/qml_time_evol_research/bug_report.py", line 231, in run_circuit_on_simulator
    result = estimator.run(qc, observable, params).result()
  File "/Users/username/venv/qu/lib/python3.10/site-packages/qiskit/primitives/primitive_job.py", line 50, in
    return self._future.result()
  File "/Library/Frameworks/Python.framework/Versions/3.10/lib/python3.10/concurrent/futures/_base.py", line 4
    return self.__get_result()
  File "/Library/Frameworks/Python.framework/Versions/3.10/lib/python3.10/concurrent/futures/_base.py", line 4
    raise self._exception
  File "/Library/Frameworks/Python.framework/Versions/3.10/lib/python3.10/concurrent/futures/thread.py", line
    result = self.fn(*self.args, **self.kwargs)
  File "/Users/username/venv/qu/lib/python3.10/site-packages/qiskit_aer/primitives/estimator.py", line 119, in
    return self._compute(circuits, observables, parameter_values, run_options)
  File "/Users/username/venv/qu/lib/python3.10/site-packages/qiskit_aer/primitives/estimator.py", line 222, in
    .result()
  File "/Users/username/venv/qu/lib/python3.10/site-packages/qiskit_aer/jobs/utils.py", line 41, in _wrapper
    return func(self, *args, **kwargs)
  File "/Users/username/venv/qu/lib/python3.10/site-packages/qiskit_aer/jobs/aerjob.py", line 106, in result
    return self._future.result(timeout=timeout)
  File "/Library/Frameworks/Python.framework/Versions/3.10/lib/python3.10/concurrent/futures/_base.py", line 4
    return self.__get_result()
  File "/Library/Frameworks/Python.framework/Versions/3.10/lib/python3.10/concurrent/futures/_base.py", line 4
    raise self._exception
  File "/Library/Frameworks/Python.framework/Versions/3.10/lib/python3.10/concurrent/futures/thread.py", line
    result = self.fn(*self.args, **self.kwargs)
  File "/Users/username/venv/qu/lib/python3.10/site-packages/qiskit_aer/backends/aerbackend.py", line 437, in
    aer_circuits = assemble_circuits(circuits)
  File "/Users/username/venv/qu/lib/python3.10/site-packages/qiskit_aer/backends/aer_compiler.py", line 552, in
    return [assemble_circuit(circuit) for circuit in circuits]
  File "/Users/username/venv/qu/lib/python3.10/site-packages/qiskit_aer/backends/aer_compiler.py", line 552, in
    return [assemble_circuit(circuit) for circuit in circuits]
  File "/Users/username/venv/qu/lib/python3.10/site-packages/qiskit_aer/backends/aer_compiler.py", line 376, in
    global_phase = float(circuit.global_phase)
  File "/Users/username/venv/qu/lib/python3.10/site-packages/qiskit/circuit/parameterexpression.py", line 475,
    raise TypeError(
TypeError: ParameterExpression with unbound parameters ({ParameterVectorElement(r[0]), ParameterVectorElement(

```

 A-tA-v added **bug** on Apr 26, 2023



ikkoham on May 2, 2023

Collaborator ...

Thank you for your report. This is a bug. Latest Aer cannot handle parameters in the global phase.
Related to: [#1781](#)



 hhorii mentioned this on May 9, 2023

🔗 [Sampler and Estimator with noise model doesn't work consistently for parametrized circuits #1798](#)

 hhorii added this to the [Aer 0.12.1](#) milestone on May 16, 2023

 hhorii mentioned this on May 17, 2023

🔗 [Support parameterized global phase #1814](#)



hhorii on May 31, 2023

Collaborator ...

[#1814](#) closes this issue.



🔒 hhorii closed this as **completed** on May 31, 2023

Support parameterized global phase #1814

<> Code

Merged hhorii merged 5 commits into [Qiskit:main](#) from [hhorii:param_for_gp](#) on May 31, 2023

Conversation 4 Commits 5 Checks 0 Files changed 8 +113 -51



hhorii commented on May 17, 2023

Collaborator

Summary

Though `ParameterExpression` can be set to `QuantumCircuit.global_phase`, Aer does not bind parameter values to it in simulation phase. This PR fixes this problem by resolving values of `global_phase` with specified `parameter_binds` in `AerSimulator.run` as well as for parameter values of gates.

Details and comments

This PR resolves [#1798](#), [#1781](#), and [#1795](#).



Support parameterized global phase

2c366a8

hhorii requested review from jakelishman and doichanj 2 years ago



doichanj reviewed on May 18, 2023

View reviewed changes

src/controllers/controller_execute.hpp

```
... @@ -65,7 +65,7 @@ Result controller_execute(std::vector<Circuit> &input_circs,
65 65 // i is the instruction index in the experiment
66 66 // j is the param index in the instruction
67 67 // pars = [par0, par1, ...] is a list of different parameterizations
68 - using pos_t = std::pair<uint_t, uint_t>;
68 + using pos_t = std::pair<int_t, int_t>;
```



doichanj on May 18, 2023

Collaborator

I think it is not necessary to change `uint_t` to `int_t` to point this params are global phases, and use pre-defined value such as `GLOBAL_PHASE_POS` instead of using `-1`



Reply...

Reviewers

jakelishman

doichanj

Assignees

No one assigned

Labels

[stable-backport-potential](#)

Projects

None yet

Milestone

Aer 0.12.1

Development

Successfully merging this pull request may close these issues.

☒ [Sampler and Estimator with noise model doe...](#)

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